Exact signed-sum spectral locus of a six-term three-qubit Pauli Hamiltonian

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Free-fermion methods identify interacting-looking spin Hamiltonians whose many-body energies are signed sums of single-particle energies. We classify this spectral property for the six-parameter three-qubit Hamiltonian $H = aZ_2 + bY_1X_2 + cX_1Y_2 + dZ_1Y_3 + eY_1X_3 + fZ_3$, an example whose frustration graph contains claws and even holes. For arbitrary real couplings, its eight-level spectrum is a multiset of the form $\{\pm\epsilon_1 \pm \epsilon_2 \pm \epsilon_3\}$, with three nonnegative single-particle energies, if and only if

$$cd\,(a^2 - f^2)(b^2 - e^2) = 0.$$

Thus the complete real locus is the union of six hyperplanes: $c = 0$, $d = 0$, $a = \pm f$, and $b = \pm e$. The proof combines exact characteristic-polynomial elimination with the chiral symmetry $Y_1Y_2Y_3$. The latter supplies a real sufficiency argument and explicit nonnegative energies, including at zero and degenerate strata. This is a spectral classification only: it does not assert a uniform Pauli-algebra construction of canonical fermion operators on the six branches.

I. INTRODUCTION

Mappings from quantum spin Hamiltonians to noninteracting fermions can expose exact spectra even when the required fermion operators are nonlinear and nonlocal in the original spins. A useful organizing object is the *frustration graph*: its vertices are Pauli terms and its edges record anticommutation. Graph conditions such as line-graph structure and the absence of claws or even holes lead to generic free-fermion constructions [1, 2]. Other constructions reach special spin-chain families outside the elementary Jordan–Wigner setting [3, 4], while twin-collapse methods enlarge the class identifiable from graph reductions [5].

The distinction between a free-fermion *spectrum* and a structural fermion mapping matters. Spectrally, one asks whether all many-body energies can be written as signed sums of nonnegative one-particle energies. Structurally, one additionally asks for canonical annihilation operators whose commutators reconstruct the Hamiltonian. The former is necessary for the latter but does not, by itself, provide a useful local or Pauli-algebra construction.

Elman, Chapman, and Flammia introduced the concrete Hamiltonian studied below and observed numerically that its all-equal-coupling spectrum is free, although generic couplings are not [1]. Its full six-term frustration graph has both claws and even holes, so the generic graph criterion in that work does not settle its fine-tuned coupling locus. We determine the exact real *spectral* locus. The answer replaces the single all-equal line by six complete hyperplanes and gives one factorized polynomial obstruction everywhere else.

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II. MODEL AND SPECTRAL QUESTION

On $(\mathbb{C}^2)^{\otimes 3}$, let X, Y, Z denote the standard Pauli matrices and let a subscript specify the tensor factor. We consider

$$H = aZ_2 + bY_1X_2 + cX_1Y_2 + dZ_1Y_3 + eY_1X_3 + fZ_3, \quad (1)$$

with $(a, b, c, d, e, f) \in \mathbb{R}^6$. The Pauli matrices are dimensionless, while all six couplings have one common energy unit. Equation (1) is a closed Hermitian 8×8 model; no thermodynamic, continuum, boundary, or perturbative limit is taken.

For $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \epsilon_3) \in \mathbb{R}^3_{\geq 0}$, define the multiset

$$\Sigma(\boldsymbol{\epsilon}) = \big\{ s_1\epsilon_1 + s_2\epsilon_2 + s_3\epsilon_3 : (s_1, s_2, s_3) \in \{-1, +1\}^3 \big\}_{\text{mult}}. \quad (2)$$

Zero ϵ_j are permitted. We seek all couplings for which $\mathrm{Spec}_{\text{mult}}\, H = \Sigma(\boldsymbol{\epsilon})$. Although a general definition may allow a common spectral shift, every term in Eq. (1) is traceless, so that shift is necessarily zero.

The six Pauli words in the order (a, b, c, d, e, f) are

$$IZI, \quad YXI, \quad XYI, \quad ZIY, \quad YIX, \quad IIZ. \quad (3)$$

The Pauli word

$$\Gamma = Y_1Y_2Y_3 \quad (4)$$

satisfies $\Gamma^2 = \mathbb{I}$ and anticommutes with all six words in Eq. (3). Hence $\{\Gamma, H\} = 0$, and Hermiticity gives

$$\mathrm{Spec}_{\text{mult}}\, H = \{\pm\lambda_1, \pm\lambda_2, \pm\lambda_3, \pm\lambda_4\}_{\text{mult}} \quad (5)$$

for ordered magnitudes $0 \leq \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \lambda_4$.

III. UNIVERSAL SIGNED-SUM POLYNOMIAL

Put $t_j = \epsilon_j^2$, and let

$$S_1 = \sum_j t_j, \qquad S_2 = \sum_{j<k} t_j t_k, \qquad S_3 = t_1t_2t_3. \quad (6)$$

Directly multiplying the eight factors associated with Eq. (2) gives

$$\chi_{\rm ss}(E) = E^8 - 4S_1E^6 + (6S_1^2 - 8S_2)E^4 - 4(S_1^3 - 4S_1S_2 + 16S_3)E^2 + (S_1^2 - 4S_2)^2. \tag{7}$$

This identity is purely algebraic and does not yet guarantee that values inferred from another polynomial are elementary symmetric functions of three nonnegative real numbers. That real-admissibility issue is handled in Sec. V.

IV. EXACT COUPLING LOCUS

Multiplication in the three-qubit Pauli group, followed by Newton identities for the traces of H^k, gives the first three even coefficients of $\chi_H(E)$ in exactly the form of Eq. (7), with

$$S_1 = a^2 + b^2 + c^2 + d^2 + e^2 + f^2, \tag{8}$$

$$S_2 = a^2d^2 + a^2e^2 + a^2f^2 + b^2c^2 + b^2e^2 + b^2f^2 + c^2f^2 + d^2e^2, \tag{9}$$

$$S_3 = a^2d^2e^2 + b^2c^2f^2. \tag{10}$$

The constant coefficient is

$$q_0 = (S_1^2 - 4S_2)^2 - 64c^2d^2(a-f)(a+f)(b-e)(b+e). \tag{11}$$

Equations (8)–(11) can also be verified by a direct exact 8×8 determinant. No coupling or energy gap is divided out.

Theorem 1. *For the Hamiltonian in Eq.* (1)*, there exist* $\epsilon_1, \epsilon_2, \epsilon_3 \geq 0$ *such that* $\mathrm{Spec}_{\rm mult}\, H = \Sigma(\boldsymbol{\epsilon})$ *if and only if*

$$cd\,(a^2 - f^2)(b^2 - e^2) = 0. \tag{12}$$

Equivalently, the complete real coupling locus is

$$\{c=0\} \cup \{d=0\} \cup \{a=f\} \cup \{a=-f\} \cup \{b=e\} \cup \{b=-e\}. \tag{13}$$

No additional inequalities are required.

Necessity. If the spectrum is a signed-sum multiset, its characteristic polynomial has the constant coefficient in Eq. (7). Subtracting it from Eq. (11) yields the exact obstruction

$$-64c^2d^2(a-f)(a+f)(b-e)(b+e), \tag{14}$$

which must vanish. Over the real numbers this is equivalent to Eq. (12). □

Figure 1 shows two coordinate slices of the four matched-coupling components. The remaining two components, $c = 0$ and $d = 0$, are coordinate hyperplanes independent of the displayed pairs.

V. REAL SUFFICIENCY AND EXPLICIT ENERGIES

It remains to show that a vanishing obstruction cannot hide complex or negative solutions for the inferred t_j. The chiral pairing in Eq. (5) provides a direct proof.

Sufficiency. Let $y_i = \lambda_i^2$,

$$U_1 = \sum_{i=1}^{4} y_i, \qquad U_2 = \sum_{1 \leq i < j \leq 4} y_i y_j, \qquad Q = 4U_2 - U_1^2. \tag{15}$$

Comparing the E^6 and E^4 coefficients of χ_H with Eq. (7) gives

$$S_1 = \frac{U_1}{4}, \qquad S_1^2 - 4S_2 = \frac{Q}{8}. \tag{16}$$

On the locus (12), equality of the constant coefficients becomes

$$64(\lambda_1\lambda_2\lambda_3\lambda_4)^2 - Q^2 = 0. \tag{17}$$

Its left-hand side factors exactly as

$$-\prod_{\sigma_2,\sigma_3,\sigma_4 \in \{-1,+1\}} (\lambda_1 + \sigma_2\lambda_2 + \sigma_3\lambda_3 + \sigma_4\lambda_4). \tag{18}$$

With the stated ordering of the four magnitudes, a zero factor is equivalent to one of the two relations

$$\lambda_4 = \lambda_1 + \lambda_2 + \lambda_3, \tag{19}$$

$$\lambda_4 + \lambda_1 = \lambda_2 + \lambda_3. \tag{20}$$

Indeed, after fixing the overall sign so that λ_4 is positive, the remaining nontrivial partitions besides Eqs. (19) and (20) contain either $(\lambda_4 - \lambda_3) + (\lambda_2 - \lambda_1)$ or $(\lambda_4 - \lambda_2) + (\lambda_3 - \lambda_1)$. Equality forces paired coincidences and hence a subcase of Eq. (20). The partitions with three or four positive terms can vanish only in the corresponding zero subcases.

For Eq. (19), choose

$$(\epsilon_1, \epsilon_2, \epsilon_3) = \frac{1}{2}(\lambda_2 + \lambda_3, \lambda_1 + \lambda_3, \lambda_1 + \lambda_2). \tag{21}$$

The four magnitudes obtained from the signed sums are $\lambda_4, \lambda_3, \lambda_2, \lambda_1$. For Eq. (20), choose

$$(\epsilon_1, \epsilon_2, \epsilon_3) = \frac{1}{2}(\lambda_4 + \lambda_1, \lambda_4 - \lambda_2, \lambda_4 - \lambda_3). \tag{22}$$

Ordering makes every entry nonnegative, and the four signed sums modulo overall sign again have magnitudes $\lambda_4, \lambda_3, \lambda_2, \lambda_1$. Equations (21) and (22) remain valid when magnitudes vanish or coincide. They prove sufficiency in Theorem 1 without assuming positivity of roots of an inferred cubic. □

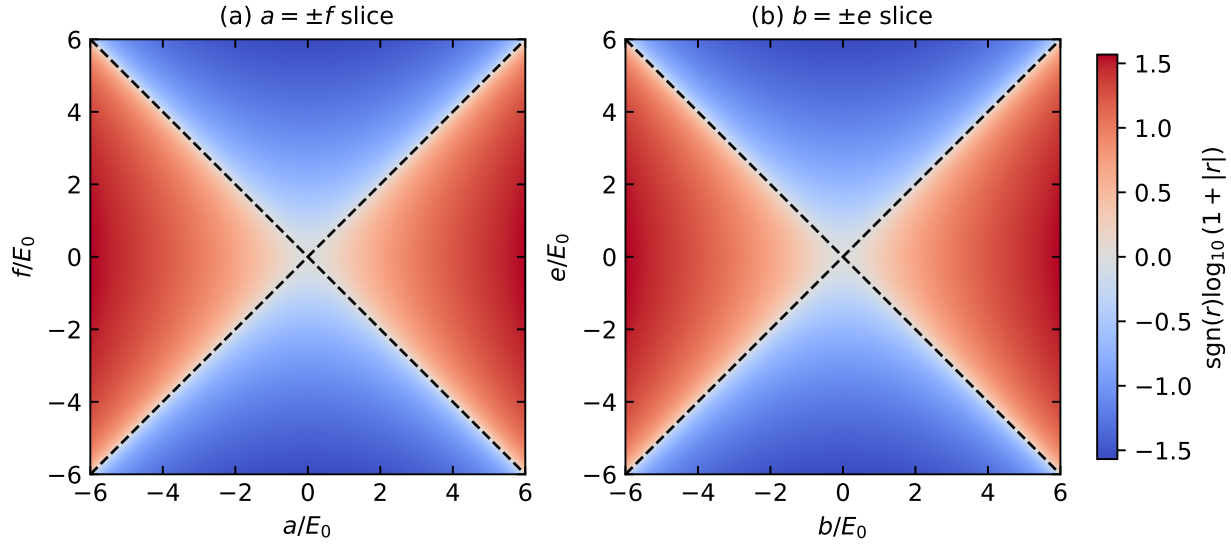


FIG. 1. Two exact coordinate slices of the factorized obstruction in Eq. (14). In (a), $(b, c, d, e) = (2, 3, 4, 5)E_0$, so none of the omitted factors vanishes and the zero set is precisely $a = \pm f$. In (b), $(a, c, d, f) = (1, 3, 4, 6)E_0$, and the zero set is $b = \pm e$. Color shows the signed monotone transform $\mathrm{sgn}(r)\log_{10}(1+|r|)$, with $r = (a^2 - f^2)/E_0^2$ or $r = (b^2 - e^2)/E_0^2$; black dashed lines are the exact zero factors. The plot is illustrative only; the classification follows from the exact factorization.

VI. BOUNDARY STRATA AND SCOPE

Because the proof never cancels a coupling, Eq. (13) includes the origin, all intersections, and all coordinate-zero strata. On the locus, a zero single-particle energy occurs exactly when $S_3 = 0$, or equivalently

$$ade = 0 \quad \text{and} \quad bcf = 0. \tag{23}$$

At least two single-particle energies vanish exactly when $S_2 = 0$; all three vanish only at the origin. Repeated squared single-particle energies are retained on the discriminant

$$\Delta_t = S_1^2 S_2^2 - 4S_2^3 - 4S_1^3 S_3 - 27S_3^2 + 18S_1 S_2 S_3 = 0, \tag{24}$$

and no part of that set is lost from the sufficiency argument.

The result should not be read as a claim of scalable free-fermion simulation. At the unrestricted finite-dimensional level, any spectrum of the form Eq. (2) permits a canonical fermion triple after choosing and labeling an eigenbasis. Such a construction is tautological and generally carries no locality or efficient Pauli representation. The nontrivial open structural question is to construct coupling-dependent canonical modes in the Pauli algebra, with controlled limits at degeneracies, especially on the four fine-tuned components $a = \pm f$ and $b = \pm e$. Theorem 1 establishes only the exact multiset condition required before such a construction can exist.

VII. PUBLISHED LIMIT AND REPRODUCIBILITY

For the published all-equal limit $a = b = c = d = e = f = J$, anticommutation of the words in Eq. (3) gives independent-set counts $(1, 6, 8, 2)$. At $J = 1$, the weighted independence-polynomial rule of Ref. [1] gives squared one-particle energies equal to the three positive roots of

$$t^3 - 6t^2 + 8t - 2 = 0. \tag{25}$$

Both the signed sums and the matrix in Eq. (1) then have characteristic polynomial

$$E^8 - 24E^6 + 152E^4 - 224E^2 + 16. \tag{26}$$

This reproduces the reported all-equal spectrum, while Theorem 1 shows that the line lies in the intersection $a = f$ and $b = e$.

The accompanying source package contains two independent Hamiltonian reconstructions: one uses Kronecker products of explicit Pauli matrices, and the other acts column by column on computational-basis bits using $Y|q\rangle = i(-1)^q|1-q\rangle$. They agree entrywise in exact symbolic arithmetic. A third implementation multiplies elements of the Pauli group, computes $\mathrm{tr}\, H^k$ through $k = 8$, and applies Newton identities. It reproduces Eqs. (8)–(11) without importing the matrix code. Scripts also verify Eqs. (18)–(22) and regenerate Fig. 1. The decisive equalities use exact integer and symbolic arithmetic; floating-point diagonalizations are retained only as diagnostics.

VIII. CONCLUSION

The free-spectral locus of the six-term Hamiltonian (1) is exactly the union of the six hyperplanes in Eq. (13). The factorization Eq. (14) excludes every off-locus real coupling, while chiral symmetry converts the same condition into a linear relation among four ordered spectral magnitudes and thereby yields nonnegative one-particle energies. The classification is exact at all zero and degenerate intersections. It sharpens a single numerical fine-tuned observation into a complete finite-model coupling-space statement, while leaving structural canonical-mode constructions as a separate problem.

ACKNOWLEDGMENTS

OpenAI Codex (model `gpt-5.6-sol`, accessed in September 2026) assisted with primary-literature discovery, symbolic-derivation exploration, code generation, independent computational checks, figure preparation, and manuscript drafting. It was directed to preserve the fixed Hamiltonian and exact real-coupling scope. Its outputs were tested through separately implemented Kronecker-product, computational-bit, and Pauli-group/Newton calculations. AI tools are not authors; final scientific and textual verification and responsibility for any submitted version belong to the author.

DATA AND CODE AVAILABILITY

No experimental data were used. All equations needed for the central result appear in the article. Exact verification and figure-generation scripts, environment versions, and regeneration instructions are included in the accompanying source package.

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